MSc PROJECT IN COMPUTER SCIENCE

PROJECT REPORT

Sales Performance Analysis and Growth Recommendations for a Super Store Using Data Analytics

Research methodology, implemented prototype and critical evaluation

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Abstract

This report examines how transaction-level data could inform sales growth and profitability decisions for a fictitious Super Store. It combines a critical review of retail analytics literature with an executable Python analytics prototype. The implementation validates transaction records, reconciles sales and profit totals, produces an offline interactive dashboard, segments customers using recency, frequency and monetary measures, and evaluates sales regression and one-step monthly forecasting against simple baselines. Twelve automated unit tests passed, and the complete analysis notebook and command-line pipeline executed on 760 clearly identified synthetic demonstration records. These results verify software operation and handling of selected edge cases; they do not establish the sales performance of the proposed store. Because the original or chosen public Superstore CSV was not supplied and independently executed, no actual dataset-based profitability claims or statistically verified business recommendations are asserted. The remaining work includes provenance-checked empirical execution, result validation, and any ethically approved human-centred testing. The technical prototype and this distinction between functional correctness and empirical validity provide the basis for the project's critical evaluation.

1. Introduction

1.1 Background and business context

Retail managers must decide which products to stock, when discounts are worthwhile, how customer purchasing patterns differ and what future demand might look like. Transaction records show what was sold, but they do not directly answer those questions. Results can also change when analysts define orders, margins or active customers differently. A dashboard focused only on revenue may celebrate products that deliver little reported profit; a narrow margin analysis may overlook the importance of repeat business. The challenge is therefore to connect valid measures to useful decisions.

The project proposal calls for an analysis of Super Store sales by product, region, customer segment and, where the data permit, sales channel. It also specifies a cleaned dataset, Python analysis, a Tableau or Power BI dashboard and ranked business recommendations (Anwer, 2026). The task here is to establish how those components should be produced and evaluated, without assuming that dashboard appearance proves research success.

Business intelligence organises stored data for monitoring; analytics investigates patterns, relationships and predictions. Total regional sales is descriptive, a proposed explanation for regional margin differences is inferential, and next-quarter demand is predictive. Each question needs different evidence. In particular, explaining historical differences is not the same task as predicting future outcomes (Shmueli, 2010).

1.2 Problem statement

The business problem is the lack of a consistent method for turning transaction history into defensible decisions. High sales in one region may conceal weak margins, while aggressive promotions may raise turnover without improving reported profit. A useful system must support both overall monitoring and detailed investigation, while allowing each headline measure to be traced back to its source records.

The central research question is: how can a reproducible retail analytics workflow identify sales performance patterns and formulate defensible, prioritised growth recommendations for a Super Store? Four subsidiary questions structure the work. First, which products, categories, customer groups, locations and time periods contribute to revenue and reported profit? Second, where do revenue growth and profit performance diverge? Third, can customer purchasing behaviour be represented by interpretable segments? Fourth, can a reasonably simple forecasting or regression approach provide useful incremental information beyond descriptive baselines? The formulation deliberately avoids presuming that sophisticated models must improve decisions.

1.3 Aim and objectives

The aim is to design and critically evaluate a retail analytics workflow that informs growth decisions without hiding profitability risks. Its objectives are to define reliable measures, prepare transaction data transparently, examine geographic and product patterns, assess customer segmentation and prediction methods, design an interpretable dashboard and connect supported findings to practical recommendations. Missing costs, confounding factors and the limitations of public demonstration data must be acknowledged.

Success requires evidence, not just software output. Data quality checks should establish missingness, identifier uniqueness and financial reconciliation. Segments need interpretable profiles and reasonable stability. Forecasts need held-out comparisons with simple baselines. Dashboard measures should be traceable to the prepared data, and recommendations should address commercial feasibility and possible unintended consequences.

1.4 Scope and boundaries

The project is deliberately restricted to transaction-oriented sales performance. It considers recorded sales, reported profit, discount levels, products, order dates, customer identifiers or segments, and geographic fields where present. It does not assume access to purchase costs by supplier, stock-out events, marketing exposure, competitor prices, employee rosters, customer satisfaction or operational capacity. The omission is consequential: without those variables, an observed association between discount and profit cannot prove what would happen if discount policy changed. A manager may nevertheless use the pattern to target investigation and controlled experiments.

A public Superstore sample can demonstrate an analytical workflow, but its transactions cannot be assumed to describe a particular operating store. Results derived from such a file need its exact version and provenance. Independent published analyses can inform the choice of questions; they are not substitutes for this project's own computed outputs.

2. Critical literature review

2.1 From raw transactions to actionable knowledge

Fayyad, Piatetsky-Shapiro and Smyth (1996) argue that knowledge discovery involves the larger process within which algorithms are only one stage. Selection, preprocessing, transformation, interpretation and evaluation are central to the usefulness of results. This is particularly relevant to retail projects, where a model with low prediction error can still be misleading if the sales ledger contains inconsistent units, repeated lines or poorly specified business metrics. Their process perspective helps prevent the analysis from becoming a sequence of unrelated charts.

CRISP-DM formalises a similarly broad lifecycle, traditionally described through business understanding, data understanding, data preparation, modelling, evaluation and deployment (Chapman et al., 2000). Its principal strength here is the explicit return from model evaluation to business understanding: an analytically valid customer segment can still lack a usable marketing action. Its weakness is that generic phases alone do not establish which tests are sufficient for a retail decision. Consequently, the project uses CRISP-DM as an organisational framework and supplements it with defined acceptance tests, change records and commercial review criteria.

Provost and Fawcett (2013) view data science through the lens of business decisions, emphasising the relationship between modelling, expected value and deployable action. Their account offers an important corrective to purely technical reporting. A hypothetical model may correctly rank high-revenue orders but be useless for campaign targeting if that revenue has already occurred or if the expected incremental margin is negative. In this project, each analytical activity therefore starts with a business question and ends with a proposed decision or a reason for withholding one. The approach is more defensible than choosing models merely because clustering and regression appear in the proposal.

2.2 Data quality as an analytical requirement

Retail data quality is often reduced to cleaning missing values, but Wang and Strong (1996) demonstrate that users understand quality more broadly. Their framework distinguishes intrinsic, contextual, representational and accessibility characteristics. For a store manager, a numerically correct revenue chart can still have poor contextual quality if it ignores cancellations or uses a time period that is not relevant to the decision. Similarly, a correct summary that cannot be traced to particular orders has limited auditability. Quality should therefore be treated as a chain from source to interpretation, not merely as the absence of null cells.

The most basic control is an explicit grain definition. In the commonly circulated Sample Superstore format, individual records represent order lines rather than distinct customer orders. If every row is counted as a separate order, order counts, average order value and purchase frequency will be inflated. A distinct order identifier is essential when calculating order-level measures, and a customer identifier should be used when calculating customer-level measures. If the source contains a row identifier, its uniqueness may provide an additional integrity check, but uniqueness must be tested rather than presumed. This example shows why business semantics are as important as data types.

Missing observations require contextual treatment. A missing postal code is not necessarily evidence of a missing sale, whereas a missing numeric Sales or Profit value prevents straightforward financial aggregation. A plausible protocol logs the percentage missing by field, tests whether omissions are concentrated by region or year and documents any repairs. Converting an unknown monetary value to

zero could materially bias profit. Similarly, removing an unusual negative-profit transaction as an outlier risks eliminating precisely the loss-making records that the project aims to explain. Exceptions should be inspected and retained unless a justified data-quality defect is found.

Data provenance is essential. The proposal mentions Kaggle Super Store data without identifying a version or supplying a source file (Anwer, 2026). Public repositories contain several variants with similar names. Their fields, dates and monetary totals must not be assumed identical. A documented dataset identifier, source location and checksum are needed before reproducible analysis can be claimed.

2.3 Measuring sales, profit and growth

At first sight, retail performance measurement appears uncomplicated. Sales can be summed, and profit can be aggregated by region or product category. Yet each aggregation embeds assumptions. Sales is typically the recorded transaction amount, not necessarily the cash received. Reported Profit in a public sample may be a dataset field calculated using undisclosed cost assumptions rather than a full audited net profit figure. Profit margin can be calculated as aggregate profit divided by aggregate sales; averaging individual transaction margins will usually produce a different number because small and large transactions receive equal weight. The aggregate ratio is more appropriate for evaluating the contribution of a business unit.

Growth must be handled with equal care. A comparison of monthly totals can be affected by the number of trading days, seasonal peaks, incomplete months and changing product assortment. Year-on-year comparisons usually provide a better seasonal reference than adjacent-month comparisons, but they still cannot establish causal drivers. A growth dashboard should present the observation window, count of transactions and category mix alongside total change. It should also distinguish monetary growth from improvements in profitability. Growing a low-margin category may increase turnover and pressure working capital without delivering the intended economic benefit.

Discount analysis is especially vulnerable to overstatement. Discount is often correlated with a reduced margin in sample transaction data, but the association can reflect product mix, region, clearance activity or historically unobserved pricing rules. For example, a heavily discounted category might have high underlying acquisition costs, while full-price categories might differ in quality or demand. A sensible analysis groups discount levels, calculates sales-weighted margin and shows sample sizes; it then compares categories and regions rather than claiming that every discounted item is loss-making. Policy recommendations should be framed as hypotheses for controlled review unless experimental evidence is available.

Observational differences are not causal effects. A margin gap between two regions warrants investigation, but does not prove that geography caused it. Similarly, discount-related losses may

reflect product mix or previous pricing decisions. Recommendations should therefore distinguish patterns in the data from the effect of changing a policy.

2.4 Customer segmentation and behavioural value

Customer segmentation seeks useful heterogeneity rather than arbitrary groups. Traditional retail segments may be demographic or organisational, whereas transaction-level datasets more naturally support behavioural segmentation. Recency, frequency and monetary value (RFM) measure how recently a customer bought, how often the customer purchased and how much the customer spent. Fader, Hardie and Lee (2005) demonstrate a connection between RFM information and lifetime-value reasoning, but the connection is not equivalent to saying that high historical spending predicts high future profit. Their work also shows why a probabilistic customer-value model requires additional behavioural assumptions.

In a Superstore setting, RFM can be constructed from unique customer identifiers, distinct order identifiers, order dates and monetary totals. Recency should be defined relative to a fixed analysis date, not the date a report happens to be opened, otherwise reproducibility suffers. Frequency should count orders rather than lines. Monetary value could represent revenue, but a parallel profit measure is valuable because two customers with identical spending can make substantially different contributions. An analyst should inspect whether very large institutional purchases dominate the scale and whether customer histories are long enough to support reliable patterns.

Clustering provides another way to represent variation. K-means groups observations according to a distance-based objective, making it computationally convenient and relatively easy to communicate. The approach nevertheless assumes that the selected scaled dimensions are meaningful and that approximately compact group structure exists. Cluster outputs are sensitive to scaling, unusual customers and the chosen number of clusters. An attractive colour-coded chart cannot establish that the groups are real or useful. Alternatives include quantile-based RFM rules, hierarchical clustering or no segmentation at all if results are unstable.

Quratulain-qurat97 (n.d.) Sample - Superstore.csv [data file]. GitHub. Available at: https://github.com/Quratulain-qurat97/superstore-retail-analysis/blob/main/Sample%20-%20Superstore.csv (Accessed: 12 October 2026).

Rousseeuw (1987) introduced silhouette analysis as a way to examine cohesion and separation in clustering. Silhouette scores provide a useful diagnostic but are not a business-value measure; a high score may identify groups that differ mainly in spending volume without suggesting distinct interventions. This project therefore proposes a multi-part selection criterion: acceptable cluster stability, interpretable RFM profiles, sufficient group size and an actionable difference in purchasing

behaviour. Cluster labels such as 'loyal' or 'at risk' should be assigned cautiously because limited transactions do not disclose a customer's intentions.

A critical issue is sampling bias. Customers appearing only near the end of an observation window have fewer opportunities to make repeat purchases; their apparently low frequency may result from limited exposure rather than disengagement. Customer-level analysis should either account for the length of observable customer history or explicitly identify the limitation. Likewise, monetary ranking based on recorded sales misses profitability and return costs. These restrictions argue for comparing simple RFM categories with algorithmic clustering rather than presuming that clustering is inherently superior.

2.5 Predictive analytics: explanatory versus predictive tasks

Regression is appropriate only after a clear target and prediction time have been defined. Shmueli (2010) distinguishes explanatory analysis from predictive modelling. An explanation asks why an outcome varies, usually with attention to coefficients, assumptions and confounding. Prediction asks whether an estimated outcome generalises to cases not used during fitting. A model that describes historical Sales as a function of contemporaneous Profit, for example, may achieve a convincing fit but have little forward-looking use because future Profit is unknown when Sales must be forecast. The feature set must follow the proposed decision point.

For a small retail dataset, a transparent baseline is often more valuable than an unnecessarily complex model. A naive forecast using last month's sales or the same month in the previous year establishes whether additional model complexity is justified. Linear regression can serve as an interpretable benchmark when relationships are approximately additive. Tree-based models can represent non-linear interactions, but may overfit sparse groups. Breiman's (2001) random forest approach reduces variability by aggregating trees and supplies a plausible comparison, although predictive importance measures must not be treated as causal explanations.

Predictive quality also depends on the evaluation design. Randomly distributing historical observations between training and test sets can leak future conditions into a model intended for forecasting. For time-indexed demand, training should precede validation, and the final evaluation period should be reserved until the modelling choices are fixed. Preprocessing steps that learn parameters, including imputation and scaling, must be fitted on training data only (scikit-learn developers, n.d.). Where the question concerns new customers rather than future months, a customer-level split might be more suitable. The experimental design must match the expected deployment context.

Hyndman and Koehler (2006) examine forecast error measures and recommend careful choice of accuracy criteria. Mean absolute error is accessible in currency units; root mean squared error penalises larger misses; and a scaled error can help compare different series. Percentage errors become problematic when actual values are zero or close to zero. Reporting several complementary

metrics and comparing them with a naive baseline is more informative than a single apparently impressive accuracy percentage. Negative results are still valid: if a simple baseline performs as well as a complex model, simplicity may be the better operational choice.

2.6 Business intelligence, dashboards and cognitive design

Few (2013) describes a dashboard as a tool for at-a-glance monitoring, with emphasis on selecting information that supports a defined purpose. A dashboard should not attempt to show every available dimension simultaneously. For the store owner, revenue, reported profit, margin, orders and growth are a reasonable top-level set; lower-level views should explain differences by region, segment, category and discount. Filters must preserve a consistent interpretation of these measures. The central challenge is not adding visualisations but selecting a small number of comparisons that support decisions.

Cleveland and McGill (1984) provide experimental evidence that graphical representations differ in how accurately observers can decode values. Position on a common scale supports more reliable quantitative comparison than decorative representations based on area or angle. This finding supports the use of sorted bars for category performance, lines for temporal variation and straightforward numerical cards for key indicators. Heavy use of pie charts, colour gradients or three-dimensional shapes would create visual complexity without improving the decision. Clear axis labels and units are not cosmetic: they form part of accurate communication.

A dashboard also contains implicit assumptions about user behaviour. A manager may need to move from an unexpected margin decline to the relevant region, then to the sub-category and finally to the underlying transactions. That navigation can be supported by consistent filters, cross-highlighting and simple tooltips. The design should include visible reset controls and clarify when an individual chart is affected by a filter. Important actions such as a proposed discount restriction should not be generated automatically from a single association; the display should link to the evidence and allow managers to inspect exceptions.

Tableau and Power BI both support the required types of visual analysis, while Python can supply reproducible cleaning, statistical testing and modelling. The choice of interface is secondary to the quality of the measure definitions and reproducibility. A screenshot is insufficient evidence that a dashboard is interactive or correct; independent totals, filter testing and representative user tasks are required. In a constrained academic project, it is better to demonstrate a small set of dependable pages than to claim a production-grade enterprise system without evidence.

2.7 Research synthesis and methodological implications

Taken together, the literature suggests a coherent sequence. Fayyad et al. (1996) and Chapman et al. (2000) establish process discipline; Wang and Strong (1996) require meaningful quality checks; Provost

and Fawcett (2013) connect analytical results to decisions; Fader et al. (2005) and Rousseeuw (1987) inform segmentation; Shmueli (2010) and Hyndman and Koehler (2006) constrain predictive claims; and Few (2013) with Cleveland and McGill (1984) inform communication. No single source supplies all of these safeguards. The methodological contribution is their integration around a retail manager's decision rather than around model novelty.

3. Research methodology and project management

3.1 Research approach

The project uses an applied design-and-evaluation approach with an executable analytical artefact. Its linked components comprise a schema-validated transactional pipeline, derived aggregates, deterministic modelling procedures, test code, an executable notebook and a privacy-minimised HTML dashboard. Software behaviour was tested using seeded synthetic records and independently verifiable reconciliation rules. Research claims about the supermarket remain conditional on selection and execution of the actual source CSV; synthetic evidence is not used to infer retail performance. This is secondary-data analytic development, not an experiment involving randomly assigned discounts or marketing treatments.

CRISP-DM is used iteratively rather than as a one-way checklist. The business understanding stage identifies sales, margin and customer-retention questions. Data understanding inspects schema, grain, coverage and anomalies. Preparation defines validated features and summary measures. Modelling evaluates exploratory analyses, RFM categories and suitable regression baselines. Evaluation checks statistical behaviour, business interpretation and usability. Deployment is restricted to a reproducible project deliverable unless access to an operational data system is separately established. If a required field is absent, the workflow should return to the business objective rather than silently substituting an unsupported measure.

3.2 Data source selection and provenance

The proposal identifies a Kaggle Super Store dataset and Python, pandas, matplotlib, scikit-learn, Jupyter Notebook and Tableau Public as intended resources (Anwer, 2026). Tableau Public describes the Superstore Sales sample as transactions from a fictitious company (Tableau, n.d.), and a public GitHub mirror exposes the customary fields required by this code (Quratulain-qurat97, n.d.). Neither the exact chosen source CSV nor evidence of its use was supplied. Instead, a separate seeded synthetic fixture was generated for software tests; the fixture is labelled in filenames and output metadata, and it is not presented as a retail research sample.

Before empirical interpretation, the full source CSV must be archived with its exact URL, retrieval date, SHA-256 checksum, data licence and date range. The implemented pipeline computes a SHA-256

fingerprint, records row counts and missing values, and raises explicit exceptions if essential columns or valid dates are absent. Returns and People tables, if sourced later, must be joined on verified keys and must not be double-counted as transaction lines.

3.3 Unit of analysis and operational definitions

Table 1. Operational metrics used in the proposed analytical workflow

Measure	Definition	Interpretation
Sales	Sum of eligible transaction-line Sales	Revenue / turnover recorded
Reported profit	Sum of line-level Profit	Dataset-specific profit measure
Margin	Sum Profit divided by Sum Sales	Weighted financial contribution
Orders	Distinct count of Order ID	Unique customer transactions
Order value	Sum Sales divided by distinct Orders	Average order size

The primary unit is the order line, identified by a row key where available. Unique Order IDs define customer orders, and unique Customer IDs define customers. A customer can place multiple orders, while each order may contain several items. This hierarchy matters whenever observations are aggregated or split for validation. Product-level analysis uses Product ID and Sub-Category; geographic analysis uses Region or State; temporal analysis uses the parsed Order Date, not Ship Date, because the commercial decision concerns the timing of a sale. Ship Date may still support a separate operational lead-time indicator if records are valid.

3.4 Data preparation and validation

Preparation begins with an unchanged copy of the source. Initial checks should record dimensions, types, dates, missing values, identifier duplication and out-of-range values. Dates need a declared format, and conversion errors should be logged rather than replaced with zero. Repeated customer or product identifiers are expected; apparent duplicate order lines must be investigated before removal.

The next stage establishes business rules. Sales and Quantity should normally be non-negative in the commonly used sample, but negative entries could also represent legitimate returns in other retail data. Those records must be interpreted according to source documentation. Discounts should be within the stated data-domain range, while unusual but valid high discounts should be retained and analysed. Where Profit is negative, that is a potential business finding rather than an automatic data error. If postal codes are used, treating them as strings preserves leading zeroes. Any missing value repair should be traceable to a rule and should not create an illusion of greater certainty.

Feature engineering is constrained by availability at decision time. Examples include order year, month, quarter, days to ship, a revenue-weighted margin measure, customer recency and order frequency. The last two require a fixed cutoff date and a clear period of observation. Customer-level features should be computed using historical data only when fed to a predictive model. A 'profit band' may

support exploratory grouping but should not appear as a predictor of Profit itself. This safeguard addresses a common source of leakage in demonstrations where targets and closely derived variables are mixed.

The prepared data should pass automated tests: unchanged eligible row count unless exclusions are documented; successful type conversions; expected identifier cardinalities; no impossible order-to-ship chronology; reconciliation of grouped Sales and Profit to the cleaned transactional totals; and stable results when the notebook is run again from a clean environment. The output should include an exception log recording the number and justification of corrected or omitted records. Without these checks, charts may be repeatable but not trustworthy.

3.5 Analytical work packages

The descriptive work package calculates key indicators for the whole period and for individual years. It examines monthly patterns, category and sub-category performance, regional comparisons, customer segments and discount distribution. Each view is designed to test a question. For instance, comparing revenue and profit side by side can identify high-turnover, weak-margin categories; plotting monthly values can reveal seasonal concentration; and displaying margin by discount band and category can help target a pricing review. The analysis must be accompanied by denominators and sample sizes, particularly in small geographic or product groups.

The segmentation package begins with transparent customer-level RFM features. Their skewness is inspected before any transformation. Where appropriate, frequency and monetary fields may be log-transformed, then standardised for clustering. A simple quantile-based grouping provides a managerial baseline. K-means may be compared for a restrained range of cluster counts, with silhouette scores supplemented by stability checks and interpretable group profiles. Labels should follow inspection of the results rather than being assumed before modelling. If groups are not stable or actionable, RFM score bands remain an acceptable output.

The predictive work package requires a choice between two distinct tasks. Aggregate monthly forecasting predicts future sales from preceding months; row-level sales regression estimates an outcome for a transaction based on inputs known at the prediction stage. They should not be mixed without explanation. For a short multi-year monthly history, a seasonal-naive forecast may be more credible than high-dimensional machine learning. If transaction-level regression is attempted, the feature set must exclude Profit and other contemporaneous outcomes that would make prediction circular. Model evaluation must include a simple baseline, temporally appropriate validation and an honest discussion of practical utility.

3.6 Tools and reproducibility

Python and pandas provide explicit data transformation and aggregation; NumPy and plotting libraries support analysis, while scikit-learn supplies standardised modelling and evaluation tools. Jupyter notebooks can document exploration, but reusable checks should sit in functions rather than duplicated cells. Tableau Public or Power BI can communicate the results when hosting and data-sharing permissions are understood. Public dashboards must not reveal private customer information.

The project should keep separate folders for original and prepared data, notebooks, tests, dashboard files and documentation. Record package versions, random seeds and output locations. Rerunning the workflow from a clean environment should recreate the published tables and figures. This protects the source evidence and makes future updates easier to audit.

3.7 Project planning and risk management

The proposed project stages are requirements definition, data confirmation, cleaning, exploratory analysis, method comparison, dashboard construction, testing, evaluation and report writing. Each stage should have a tangible acceptance criterion. For example, the data stage concludes with a schema record and quality audit; the modelling stage concludes with a reproducible comparison against a baseline; and the dashboard stage concludes with numerical reconciliation and user-task testing. The proposal specifies weekly milestones and fortnightly progress reporting to the supervisor (Anwer, 2026). Any actual milestone completion or supervisor feedback requires dated supporting evidence and should not be inferred from the proposal alone.

3.8 Ethical considerations

The analysis may include customer names or identifiers in a public sample, but the dashboard ordinarily needs only aggregated indicators. Data minimisation principles suggest retaining or exposing only what is necessary for the analytical purpose (ICO, n.d.). Where genuinely identifiable customer information is involved, the legal basis, storage arrangements and retention period require review. Sharing extracts on public dashboards or repositories demands particular caution. If stakeholder interviews or formal user testing are undertaken, appropriate informed consent and any required university ethics approval must precede data collection.

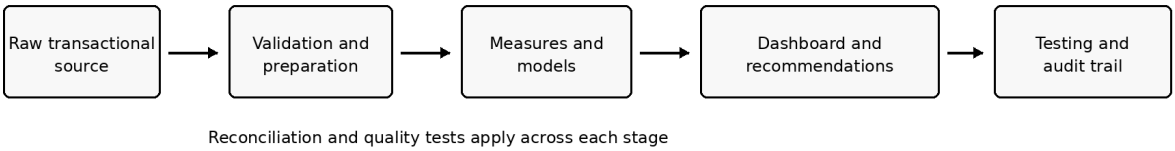
The project proposal states that an ethical approval form would be submitted before implementation (Anwer, 2026). That is a planned action, not documentary proof that approval was granted. The final report should record the true approval status and the scope of any human-participant activity. If no stakeholders were recruited, the evaluation should say so rather than inventing interview numbers or feedback. Ethical transparency is closely connected to scientific validity because unrecorded data provenance and fictitious testing can render even plausible conclusions unreliable.

4. High-level and low-level design

4.1 Architectural overview

The implemented prototype separates source input, validation, group-level measures, segmentation, prediction and dashboard construction into documented Python modules. A supplied CSV is checked for required field names, unique Row IDs, numeric bounds and chronological consistency. Cleaned records feed independently reconciled regional, category, segment and discount-band summaries, followed by customer clustering and model evaluation. A command-line interface writes the audit, metrics, selected plots and self-contained HTML dashboard into an output directory; a second entry point allows execution of an explicitly synthetic fixture for testing.

Figure 1. Implemented analytical pipeline and audit checkpoints



The design avoids tightly coupling a notebook cell to a screenshot. A dashboard value such as total profit should derive from a named measure applied consistently after the documented exclusions. The same measure should be independently reproduced by a validation script. When a user filters the dashboard to one region, the remaining charts should all show that same filtered population unless a comparison with the whole business is clearly labelled. A change in data cleaning should trigger recalculation and reevaluation rather than leaving saved graphics inconsistent with revised figures.

4.2 Data structures and analytical grain

The central prepared table retains one row per eligible order line and fields sufficient to reproduce major dimensions. A customer summary table has one row per Customer ID for a defined historical window, with recency, frequency, monetary value, profit contribution and total quantity. A monthly summary table has one row per time period and, when needed, per geographic or category group. This layered structure is preferable to repeatedly calculating every feature directly in a dashboard because it permits explicit validation at each granularity.

The data dictionary should describe each measure in business language and technical terms. 'Monthly Sales' is the sum of line Sales values grouped by Order Date calendar month. 'Order Count' uses distinct Order IDs rather than a simple row count. 'Profit Margin' is a ratio of sums, not the mean of row ratios. 'Customer Frequency' counts distinct orders before a fixed analysis cutoff. 'Days to Ship' is the date difference between Ship Date and Order Date for records with both fields present. Definitions become part of the artefact, not explanatory footnotes that can be omitted when the dashboard is published.

4.3 Dashboard layout and interaction

The delivered dashboard is a single-page, offline HTML application showing four key measures: total sales, total reported profit, weighted profit margin and sales-line count. JavaScript filters allow selection by year, region, category and customer segment. Four Plotly charts show monthly revenue, profit by region, category-level sales versus profit and the distribution of profit across discount bands. The selected sales and profit KPI totals are recalculated from the current filtered records; the underlying standalone page does not embed customer names or individual customer IDs. Its calculations are reproducible, but human usability outcomes have not been measured.

The HTML page also supports diagnostic review of weak-profit regions and discount patterns. The source date shown on the dashboard is the Order Date, while a separate audit preserves shipping-related validation. Visualisations use a limited colour palette and reserve contrasting colours for negative profit. The dashboard is generated from the selected input on each run, so no spreadsheet manually copied into the presentation layer is required.

Customer-level RFM features and clustering profiles are exported into a local internal CSV for further review; they are not published in the browser dashboard. This separation limits unnecessary disclosure of customer identifiers. More specialised drill-downs, return-rate analyses and forecast confidence intervals remain future extensions rather than documented completed features.

Few (2013) and Cleveland and McGill (1984) support the choice of a restrained layout. Horizontal bars enable accurate comparison of many category names; a small set of colours can distinguish revenue and profit rather than decorating the interface; and carefully labelled filters reduce the risk of misinterpretation. The dashboard's design priority is the speed and correctness of business interpretation, not visual novelty. Standard page titles, visible date ranges and stable number formats are required throughout.

The captured screen in Figure 2 records a functional dashboard-filter check. Its synthetic-data notice is deliberately retained, and the displayed values should not be read as observations of the actual Superstore dataset.



Figure 2. Tested dashboard with year and region filters (synthetic demonstration data only).

4.4 Algorithms and evaluation interfaces

Customer segmentation has been implemented with recency measured from the latest observed order plus one day, frequency based on distinct orders, and monetary value based on line-level sales. Log transforms and standardisation precede seeded K-means clustering; silhouette scores compare admissible k values. The checkout-stage row-sales regression uses a chronological holdout, a training-mean baseline and a Random Forest with predictors restricted to geography, segment, product category, quantity, discount and calendar features; Profit and Sales are excluded from predictors. Monthly forecasting implements rolling one-step backtests for the previous-month baseline, seasonal-naive baseline and Ridge regression using only historical lags. Both prediction tasks return MAE/RMSE and transparent diagnostics, not claims of prospective deployment performance.

4.5 Security, reliability and maintenance

The modular scripts validate input and fail on invalid schema or dates. Source checksums and dependency versions support reproducibility. Unit tests cover negative-profit lines, duplicate keys, invalid discounts, shipping chronology and reconciled totals. The offline dashboard uses aggregated values rather than customer identifiers. User deployment would still require security, usability and governance reviews.

4.6 Implemented code and reproducibility

The executable project contains src/analytics.py for schema validation, aggregations and modelling; src/dashboard.py for a standalone, responsive Plotly dashboard; src/demo_fixture.py for seeded test data; run_analysis.py as the repeatable entry point; a Jupyter Notebook; and tests/test_analytics.py. The CLI accepts --input for a genuine, provenance-recorded CSV or --demo for an intentionally synthetic fixture. Every completed run exports a SHA-256 source fingerprint, missing-value audit, cleaned internal data, grouped summaries, customer clusters, model comparisons, charts and an offline dashboard. Running the notebook from a clean kernel recreates the same demonstration analytical steps. This code is reference implementation material: actual empirical results, manual customer research and claims of independent student authorship are not established by software execution alone.

5. Analytical specification and evidence framework

5.1 Descriptive and diagnostic enquiries

A completed analysis should start with a reconciled summary of the observation window and record counts. Reported revenue, profit, margin, orders and customers should be calculated directly from a named prepared dataset. Thereafter, each analytical comparison should answer a specific question. Category and sub-category tables identify the distribution of sales and profit. Region- and state-level measures reveal whether contributions vary geographically. Monthly plots show potential seasonality and whether revenue and margin move together. A review of discounts examines the relationship between commercial incentives and observed profitability.

5.2 Secondary benchmark observations

Public implementations of Sample Superstore provide useful benchmarks for checking whether an analysis appears plausible, but they cannot replace an independently executed investigation. One public project describes the commonly circulated version as containing 9,994 order lines, 5,009 orders and 793 customers and reports sales of approximately US$2.30 million and profit of approximately US$286,000 (Borges, n.d.). Another independently published dashboard describes strong differences between discount bands and identifies Texas and Illinois as significant contributors to negative profitability in a particular version of the dataset (thummam, n.d.). These are findings claimed in external projects, subject to their own preprocessing choices and dataset versions.

The appropriate academic use of these reports is comparative rather than imitative. The existence of a reported weak-margin region suggests a hypothesis to investigate, not proof that the dataset selected for this project has the identical problem. A disciplined study would first reproduce global counts on the chosen file, then check region and discount results, and finally explain any discrepancies. Distinct

definitions of fiscal period, returns, order grain, or data cleaning could account for differences. This benchmarking activity is useful precisely because it exposes the sensitivity of business conclusions to assumptions.

5.3 Analysis of discounting and profitability

The discount question is commercially important because it can lead to an immediately actionable recommendation. The proposed analysis first groups lines by recorded discount levels, then calculates revenue, profit, margin, quantity, number of lines and distinct orders in each band. It then repeats the comparison inside major categories and regions. This layered structure avoids attributing a weak-margin category entirely to discounting when its underlying product mix may differ. In addition, the proportion of lines with negative profit should be reported alongside total losses.

The interpretation should remain conditional. Observing lower margin in heavily discounted transactions is consistent with a pricing concern, but does not reveal demand elasticity. Removing a discount might reduce sales volume enough to offset the margin benefit, or it might redirect customers to competing products. The strongest recommendation from observational data is therefore to identify discount rules requiring managerial review and to test controlled changes. A suggested intervention might involve pre-approval for unusually deep discounts in categories repeatedly associated with losses, accompanied by tracking of volume, profit and customer outcomes.

5.4 Time-based analysis and seasonal effects

Monthly aggregation supports operational planning, but a four-year sample offers relatively few independent annual seasonal cycles. The analysis should compare month-of-year patterns across years and avoid over-interpreting a single holiday spike. Growth percentages need an explicit base period, and incomplete boundary months should be excluded or flagged. When observations are sparse in a sub-category, smoothing may clarify trends but can also hide abrupt changes. The dashboard should permit users to see unsmoothed totals or clear notes about smoothing.

5.5 Customer groups and practical targeting

An RFM table can describe customers with recent frequent purchases, infrequent large purchases, historically valuable but inactive relationships and low-volume occasional orders. These descriptions are potential targets for further investigation, not verified psychological customer types. A cluster with high revenue but weak profit contribution may warrant contract or discount review rather than an indiscriminate loyalty reward. Conversely, a modest-spending cluster with healthy margins may support low-cost retention activity. Customer targeting should be based on expected incremental profit, not only on historical spending.

The analysis should also avoid exaggerated precision. With short purchase histories, a customer who appears to be new may actually have transacted outside the source period. Monetary rankings are sensitive to rare large purchases. Analysts should test whether segments remain recognizable after excluding a small number of extreme orders and whether differences persist across time windows. If the grouping changes radically, the correct conclusion is that the segmentation is fragile. Maintaining a simple rules-based view may then provide greater managerial reliability than deploying unstable clusters.

5.6 Prioritising recommendations

Business recommendations can be prioritised along four dimensions: size of the observed issue, quality of the supporting evidence, cost or difficulty of the intervention and risk of an adverse consequence. A discount audit may be high priority when substantial negative profit is concentrated in a small set of practices, but a firm-wide discount ban would have high downside risk without experimentation. A category assortment review may be justified by persistent low profitability, though supply contracts and strategic products require consideration. Regional performance comparisons can identify where to investigate freight or promotional practice without asserting a single cause.

For each intervention, the proposed evaluation should identify an owner, a measurable outcome and a review period. For example, a narrowly scoped discount pilot could track order volume, gross reported profit, margin and returns, comparing results with a credible control group if available. A customer retention initiative could monitor repeat-order behaviour and incremental margin rather than raw campaign response alone. This requirement operationalises Provost and Fawcett's (2013) argument that analytical output should be connected to decision value.

5.7 Why unsupported numerical findings should be withheld

An executed notebook, generated graphs and an offline dashboard establish technical functionality on 760 synthetic transactions, not on an identified Superstore source. The fixture outputs cannot justify empirical business claims; dataset-specific results and user evaluation still require independent verification.

6. Testing and evaluation

6.1 Evaluation philosophy

A working program is not automatically an analytically valid or usable solution. Evaluation must separately assess source quality, calculation accuracy, predictive performance and business relevance. A testable check is needed for each risk.

Evaluation needs more than screenshots or a successful notebook run. It should explain why a method was chosen, compare credible alternatives and document failure cases. A finding that a simple baseline performs as well as a complex model is still informative.

6.2 Data and transformation tests

Initial validation should confirm that required fields exist, dates parse consistently, numerical ranges are plausible and key identifiers behave as expected. Reconciliation tests compare the total Sales and Profit before and after transformations, accounting explicitly for any exclusions. Grouped figures should sum to the same total as the ungrouped cleaned data. Derived fields should be tested on small examples whose correct answers are known in advance. For instance, a two-line order must count as one distinct order, and a customer with purchases on three dates must have a frequency of three rather than the number of purchased items.

Edge cases deserve attention. A category with zero sales should not generate an undefined displayed margin without explanation. Dates crossing month or year boundaries should be assigned consistently. A negative-profit transaction must not be dropped because it produces an inconvenient plot. Missing customer IDs may prevent customer-level analysis but need not automatically invalidate overall sales aggregates. A good test suite records both the exception and the adopted business treatment, allowing a reviewer to distinguish between a data defect and a deliberate analytic choice.

6.3 Model validation and comparison

Customer clustering should be assessed through profile interpretability, stability under reasonable perturbations and internal diagnostics such as silhouette width (Rousseeuw, 1987). A single best-looking two-dimensional scatter plot is not sufficient. The evaluation should record feature scaling, number of clusters and the proportion of customers in each group. Stability can be inspected by changing random seeds or modestly perturbing samples, while acknowledging that numerical labels can be permuted. Results should be translated into meaningful differences in recency, frequency, spend and profit contribution.

A regression or forecast component requires a clearly defined held-out evaluation period and baseline. For forecasting, rolling-origin assessment may offer a better approximation to repeated future deployment than a one-off random split. The chosen metrics should include MAE and RMSE, with a scaled metric where suitable (Hyndman and Koehler, 2006). Any model tuning should occur within training or validation data. The final period must not influence feature selection, imputation or model choice, as such leakage inflates apparent performance (scikit-learn developers, n.d.).

Model error should also be interpreted operationally. A reduction in average absolute error is useful only if the forecast arrives before a decision and produces a material benefit. A model that makes better predictions of historic line-item Sales using information available only after purchase has no

forward-looking value. Similarly, a complex model that gains a small improvement but cannot be explained or maintained may be less appropriate than a simple baseline for a small retail organisation. These criteria connect statistical testing to business utility.

6.4 Dashboard functional and usability testing

Functional tests should verify that KPI cards agree with a separately calculated reference total, year filters update all relevant visuals, category selection does not accidentally alter comparison denominators, and selecting an empty data slice produces a clear message rather than a misleading zero. Sorted categories, labels and number formats need visual review. Date filters must not mix Order Date with Ship Date without explicit explanation. Exported tables must respect intended privacy restrictions. Screenshots alone cannot establish that interactions work; a repeatable test script should list actions and expected outputs.

Usability evaluation can be organised around realistic tasks. A reviewer might be asked to identify the least profitable sub-category, find the region with the largest absolute loss, compare two years of reported profit and inspect the distribution of discount levels in a selected category. For each task, the test should record whether the answer was correct, how long it took and what confusion arose. Participants should be recruited only under applicable ethics requirements, and their number and characteristics must be reported accurately. If human testing is unavailable, an expert heuristic review can be described as such, but should not be presented as equivalent to a user study.

6.5 Evidence matrix and readiness judgment

Table 2. Evidence available and still required for empirical completion

Workstream	Minimum verifiable evidence	Evaluation question
Data	Implemented schema checks and demo audit; actual CSV not supplied	Are totals reliable?
Analysis	Executed demo summaries, graphics and reconciliation tests	Can conclusions be traced?
Models	Executed synthetic-data baseline comparisons; real-data metrics pending	Is added complexity useful?

Workstream	Minimum verifiable evidence	Evaluation question
Dashboard	Offline dashboard and browser-filter check passed; human usability still pending	Can users answer intended questions?
Recommendations	Decision logic discussed; data-backed ranking requires actual records	Are actions justified and testable?

The project now has implementation-level evidence: source code, reproducible demonstration execution, CSV reconciliation, a generated dashboard and a recorded 12-test unit suite. That evidence supports the conclusion that basic software modules operate on the synthetic test fixture. It does not validate empirical claims for a Superstore source dataset, end-user usability, commercial impact, research ethics approval or independent project execution by the candidate. Those areas should be assessed separately when suitable records are available.

6.6 Recorded implementation tests and limitations

A fresh execution of python -m unittest discover -s tests -v completed 12 of 12 tests successfully. The tests cover retention of loss-making lines; distinct order counting; total reconciliation by region and month; seeded clustering reproducibility; future-dated holdout separation for the sales regressor; rolling monthly backtesting; and rejection of invalid discounts, missing required fields, duplicate Row IDs and shipping dates earlier than their order dates. The analysis command and executed Jupyter Notebook also completed using 760 seeded synthetic order lines, generating five grouped summaries, the RFM output, both prediction comparisons, three figures, source audit JSON and a self-contained interactive HTML dashboard. These checks are evidence of functioning software on test inputs, not of numerical accuracy on the actual Superstore dataset or of successful user-task testing. A separate headless Chromium check confirmed that all four dashboard filters render, and that selecting a different region and year updates the sales KPI without JavaScript errors. This is an automated interface check, not a human usability study.

Figures 3–5 are the charts generated by the executed demonstration pipeline. They verify that the plotting and aggregation functions produce legible outputs; the trends, magnitudes and relative rankings are invented by the test fixture and are not empirical findings.

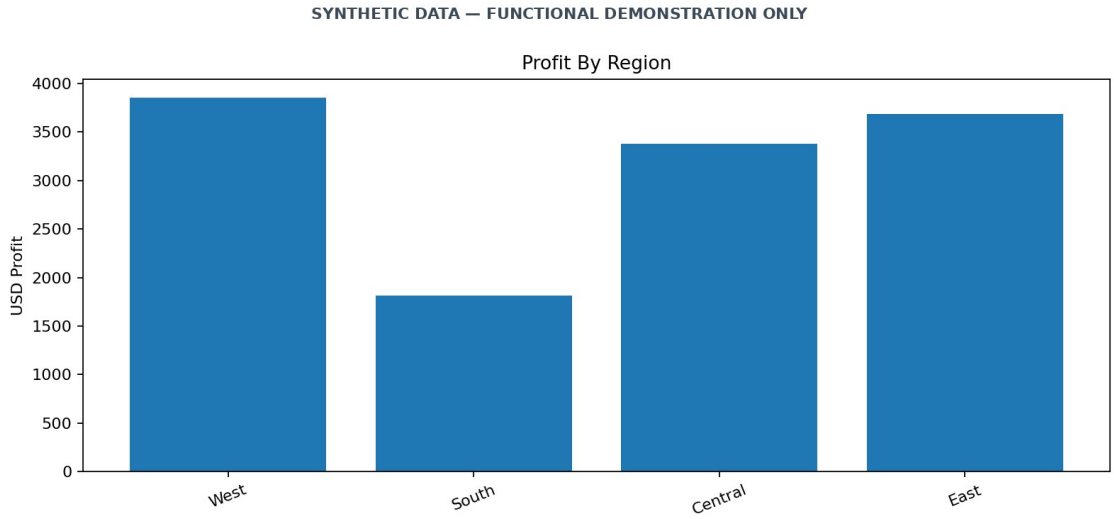


Figure 3. Region-level reported profit: synthetic execution, not observed business performance.

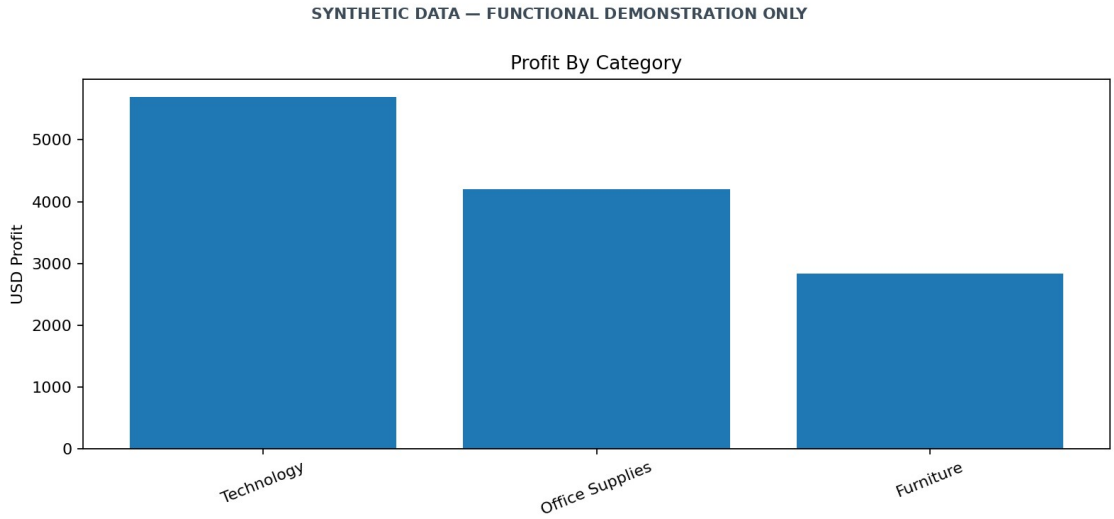


Figure 4. Category-level reported profit: synthetic execution, not observed business performance.

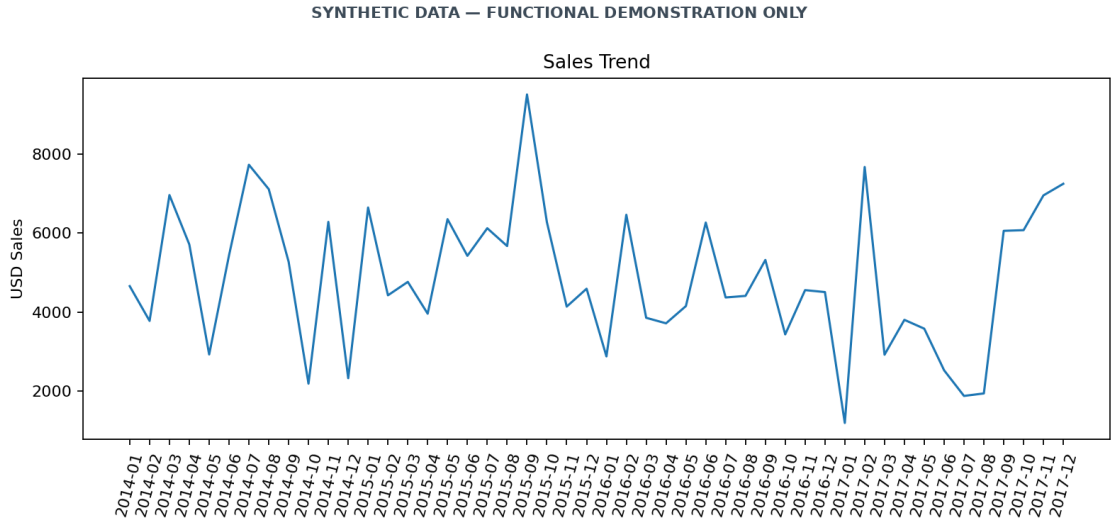


Figure 5. Monthly sales plotting: synthetic execution, not a validated Superstore trend.

7. Critical evaluation and reflection

7.1 Strengths and scope of the implemented prototype

The design's main strength is its attention to measurement. It separates turnover from reported profit and distinguishes orders, order lines and customers. Reconciliation between Python and dashboard calculations should prevent filters or calculated fields from silently changing reported totals. This supports the project's stated aim of producing usable business recommendations.

The methods are also proportionate to the data. Descriptive diagnostics can be useful without elaborate algorithms; transparent RFM categories provide a baseline for customer segmentation; and forecasting is judged against simple time-series rules. Such comparisons make it possible to explain both successful methods and those that do not justify their added complexity.

7.2 Weaknesses of transaction-only evidence

The largest limitation is the absence of important explanatory variables. A retail sales file generally lacks product acquisition costs, stock-outs, changing competitor prices, marketing exposure, service quality and broader demand conditions. Even when a Profit field is supplied, its accounting scope may be unclear. It may exclude rent, labour, financing and allocation effects. Consequently, an analysis of reported profit is not a full profitability audit. Recommendations concerning pricing and assortment should be interpreted within that financial limitation.

The second limitation concerns causal inference. In observational transactions, discounts are not assigned at random, and customers or categories receiving them may differ from those that do not. A regression coefficient on Discount cannot automatically be interpreted as the causal effect of a discount on Profit or demand. Similar uncertainty applies to regional differences and the apparent effects of customer segment. This matters because a manager could incur real losses by applying a causal interpretation to descriptive evidence. The project therefore favours controlled pilots and clearly qualified language.

Public demonstration data may be dated or unrepresentative of a real store. The recorded time period, geography and provenance constrain how far results can be generalised. Realistic customer and product fields do not establish that the file contains confidential operational records. Application to a new retailer needs independent data and validation.

7.3 Segmentation limitations and alternatives

Behavioural clustering may oversimplify customer dynamics. A snapshot of recency and frequency cannot distinguish genuine disengagement from changes in available observation time or seasonality. Large one-off business purchases may strongly influence scaled monetary features. K-means imposes a geometry on the data that may not correspond to natural customer groups. An apparently acceptable

silhouette score gives evidence of mathematical separation, not proof of different response to a marketing action. These weaknesses justify sensitivity analysis and side-by-side comparison with transparent RFM rules.

Alternative approaches should be chosen according to the business decision. If the objective is to decide which customers merit a low-cost follow-up, a simple threshold rule may be preferable. If the objective is to estimate future purchases or customer lifetime value, models inspired by Fader et al. (2005) may be conceptually better aligned, but require adequate transactional histories and assumptions about purchasing processes. If customer records are too short, no reliable segmentation may be possible. The project should treat an abstention from weak modelling as a valid scientific choice.

7.4 Prediction limitations and alternatives

The modest time horizon of common Superstore samples constrains demand forecasting. Four years of monthly values give only a few observations of each seasonal month. Complex models with many predictors risk fitting historical irregularities rather than general patterns. Prediction also becomes fragile if the period contains structural changes not represented by known features. Comparing candidate models with seasonal-naive forecasts, as suggested by time-series literature, guards against the assumption that machine learning is automatically more accurate than a simple rule.

A model's operational value depends on timing. If a retailer needs forecasts six weeks before purchasing stock, it must be trained using information available at that lead time. Models using final monthly profit or transaction-level discount decisions that are only known after the sale would not support the intended decision. This may reduce measured accuracy, but it greatly increases validity. The project consequently defines prediction timing before feature selection and prohibits evaluating the final period repeatedly while adjusting the model.

7.5 Dashboard risks and organisational adoption

A dashboard can create false certainty by presenting precise numbers without visible caveats. A single profit card may conceal uncertainties about return handling or the definition of costs. Colour choices can exaggerate differences, and complicated interactions can lead a user to compare filtered and unfiltered totals unknowingly. The design must therefore use consistent units, visible filters, source notes and drill-down explanations. A decision-support interface is effective only when a manager understands both what a number means and what it excludes.

Adoption is another uncertainty. The owner may prefer a small weekly summary to a sophisticated interactive interface, particularly if the business lacks staff to maintain data connections. Requirements should be evaluated with actual users when possible. In the absence of approved stakeholder access, usability cannot legitimately be declared successful. A design review or checklist can provide partial

evidence, but the distinction should remain visible. This is especially important because the project title promises recommendations rather than simply a visual demonstration.

7.6 Reflection on project management and independent learning

The proposal outlines a staged lifecycle and regular supervisor contact (Anwer, 2026). A final reflection, however, should compare planned and actual progress using authentic development records. If data access caused delays, for example, the report should explain which tasks were rescheduled and why. It should not invent meetings, completed milestones or stakeholder feedback.

One important methodological lesson is that each technique needs a reason to be included. Validation, clustering, regression and dashboard design address different questions and create different maintenance costs. The design prioritises traceable measures and straightforward methods. Independent learning must ultimately be evidenced through the researcher's own decisions, revisions and records.

7.7 Societal, economic and environmental considerations

Sales analysis may help reduce ineffective promotions or improve stock planning, but its wider benefits are not guaranteed. Higher margins may also affect affordability, and marketing focused only on historically valuable customers may overlook new groups. Recommendations should therefore consider fairness, privacy and the limits of transaction data.

Forecasting might reduce overstocking, but inventory and waste information would be needed to measure that benefit. The proposal also identifies possible economic and social gains, including better staffing or service if sales rise (Anwer, 2026). Those outcomes are possibilities, not demonstrated effects of the proposed dashboard.

8. Recommendations and implementation roadmap

8.1 Establish a trusted performance baseline

The first recommendation is to standardise the definitions of sales, orders, profit, margin and customer frequency and make them visible in the management system. A trusted baseline should be reconciled against the prepared transaction file and updated on a fixed schedule. Rather than beginning with machine learning, the store should initially monitor a small number of measures covering revenue, contribution and change. This is a low-complexity, comparatively low-risk intervention that enables all later analysis. Its success can be measured through reconciliation accuracy, reporting time and reduction in conflicting totals.

8.2 Investigate concentrated sources of profit leakage

A second recommendation is to review sub-categories, geographic groups and discount levels that jointly account for substantial absolute negative reported profit. The investigation should first determine whether the pattern is persistent across periods and whether large transactions dominate the result. A focused discount review may be warranted where deep promotions coincide with sustained losses, but the initial action should be diagnostic. The manager should examine buying costs, returns, freight and promotional objectives before imposing broad restrictions. A subsequent pilot should track total contribution and lost volume, not merely margin percentage.

8.3 Segment customers cautiously and test interventions

The third recommendation is to create simple behavioural customer profiles using a fixed RFM window, with profitability alongside spending. Marketing interventions should be linked to an explicit aim, such as repeat purchase or reactivation, and measured against a control or reasonable comparison group. Broad labels should not substitute for evidence of effectiveness. When clustering does not produce stable, understandable groups, rule-based RFM categories should be retained. This approach delivers segmentation benefits without requiring an unmaintainable modelling pipeline.

8.4 Use conservative forecasting for planning

Forecasting should begin with a seasonal naive baseline at monthly category level, not a complex algorithm. A proposed alternative model should be retained only when rolling-origin tests demonstrate a consistent, useful reduction in error. Where uncertainty is substantial, managers should use ranges and scenario analysis rather than a single point estimate. Forecasts should guide planning discussions, not automatically trigger inventory purchases.

8.5 Govern the dashboard and analytical lifecycle

Assign responsibility for data refreshes, reconciliations, access permissions and definitions. Each new version should retain the source file, transformation record, model settings and dated validation outputs. Quarterly reviews should ask whether measures still support real decisions and whether action taken on the recommendations has improved financial outcomes. Outdated, unexplained or unreconciled charts should be withdrawn until corrected.

9. Conclusion and future work

Retail transactions do not automatically reveal where profitable growth is possible. The executable prototype demonstrates how schema-controlled data preparation, reconciled financial measures, RFM clustering, temporally separated model comparisons and an offline interactive dashboard can be combined into a reproducible analytical system. Twelve unit tests and an end-to-end notebook

execution provide bounded evidence that these software components run on synthetic test records; they do not validate performance on an actual Superstore file.

This implementation is therefore more substantial than a design proposal but remains short of an empirical evaluation. Its outputs cannot justify quantified interventions for the supermarket until the chosen source CSV has been obtained, analysed and independently interpreted. Public third-party Superstore analyses may contextualise research questions but are not substitute evidence for the present executable project.

The necessary next step is to identify, cite and archive the exact Superstore dataset, then execute the supplied pipeline and notebook on that version and verify actual summary totals, model scores and dashboard behaviour. Additional work includes meaningful business validation, ethics-approved usability research if relevant, and independent student reflection. The report is a technical reference draft until these requirements and the university's academic integrity declaration have been satisfied.

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